Physics Transformers: Tensor Inputs, Law-Conditioned Attention, and Fused Reasoning–Physics Layers

Sehaj Randhir Singh
Department of Electrical and Computer Engineering,
NYU Tandon School of Engineering (independent researcher)
Brooklyn, NY, USA
sehajrsinghs@gmail.com

***Abstract*—We present PhysFormer, a transformer adjusted for physics, and Law-Conditioned Attention (LCA), a new way to feed physics into it. The governing equation is tokenized into a fixed symbolic vocabulary, embedded into a law vector, and injected as a cross-attention key/value stream in every layer, giving physics an input channel in addition to the usual loss channel. Across 36 paired runs, LCA reduces trajectory error by 21% (p = 0.0003); swapping the equation signature at inference causally steers the prediction (p ¡ 0.0001) while a constant-signature control is exactly insensitive. The loss channel, isolated, buys consistency (6–8× residual reduction) but not held-out accuracy. We pre-registered a regime theory – benefit should be monotone in token-vocabulary ambiguity – and falsified it on a ten-law suite (Spearman $\rho = 0.07$, p = 0.88), reporting the failure analysis rather than a swept-under null. Against ten dedicated per-law DeepONets (median held-out error 0.037), the single generalist serves all ten laws at 0.110 with no law identity at inference, beating the specialists outright on spring and LC.**

***Index Terms*—physics-informed machine learning; transformers; multi-domain learning; law-conditioned attention; neural operators**

I. Introduction

Physics-informed machine learning has converged on one template: a network fits the solutions of a single governing equation, and physics enters only as a penalty on the output. We show physics can enter as input tokens – the equation signature itself – so the network can use it at inference, not just during training. This paper measures the two channels separately (loss vs. input), tests a pre-registered theory of when conditioning pays, and compares against an external operator-network baseline.

II. Method

PhysFormer embeds each physical quantity token (length, load, stiffness, ...) and operator token into a shared vocabulary, forms a law vector from the equation signature, and injects it as cross-attention in every transformer layer with a scale-bias gate on the readout. A physics-consistency layer computes the residual of the predicted trajectory against the governing equation and adds it to the loss. The multi-law protocol trains one shared body and shared heads on all laws at once (96 samples per law, 250 epochs, 3 seeds); the control is a dummy-signature condition with identical architecture and data.

TABLE I
Pre-registered 10-law regime test: measured LCA benefit (median over three seeds) vs. equation ambiguity computed from the token vocabulary alone. The pre-registered monotone prediction is falsified ($\rho = 0.07$, $p = 0.88$).

Law	Ambiguity	Benefit
beam	1.000	+0.719
cantilever	1.000	+0.340
projectile	0.500	+0.141
pendulum	1.000	+0.539
spring	1.000	-0.407
rc	0.667	-0.049
damped	0.750	-0.013
kepler	0.667	-0.134
lc	0.667	-0.149
drag	0.500	+0.110

III. Results

Law-conditioned attention reduces trajectory error by 21% (p = 0.0003) over 36 paired runs, and the effect concentrates on the one token-identical pair (beam/cantilever). The pre-registered regime prediction failed; the failure analysis shows the overlap measure conflates token-superset relations with genuine indistinguishability. Ten dedicated DeepONets reach median error 0.037 vs. 0.110 for the single generalist – a tradeoff stated plainly, not a win claimed.

IV. Honest limitations

The pooled single-model DeepONet (law identity as an explicit one-hot branch input) did not complete under

TABLE II

EXTERNAL OPERATOR-NETWORK BASELINE. PER-LAW DEEPONET HELD-OUT TRAJECTORY ERROR VS. THE SINGLE LAW-CONDITIONED GENERALIST (MEDIAN 0.037 VS. 0.110 OVER THE NINE NON-DEGENERATE LAWS).

Law	DeepONet error
beam	0.025
cantilever	0.009
projectile	0.011
pendulum	0.037
spring	0.122
rc	0.007
damped	0.141
kepler	0.066
lc	0.192
drag	0.011

available compute and is reported as an attempt, not a result. The refined regime hypothesis remains open: pendulum shows consistent benefit with no token twin, and learned embeddings do not confound its tokens (max cosine 0.30).

REFERENCES

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